

GlobeTrack Logistics Ltd.

Big Data Engineering Project

End-to-End Cloud Big Data Solution for Real-Time Logistics Intelligence

TCS iON AIP-135

AWS ap-south-1

Python 3.12

XGBoost + LightGBM + Random Forest

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Cloud Platform: AWS Asia Pacific Mumbai ap-south-1

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GitHub: github.com/Fatema-016/globetrack-logistics

1. Executive Summary

This project implements an End-to-End Big Data Analytics pipeline for GlobeTrack Logistics Ltd. — a logistics company operating across 8 major cities in Maharashtra and Goa, India. The system collects, processes and analyses simulated data from three heterogeneous real-time sources: vehicle IoT sensors, customer booking logs and external weather and traffic APIs.

The solution is built entirely on AWS cloud infrastructure in the ap-south-1 Mumbai region.

1.8M

Raw records ingested

1.18M

Records processed

3

S3 data zones

94.5%

Fleet health

27/27

Tests passing

2. Project Objective

The primary objective is to design and implement a scalable Big Data Analytics solution on AWS cloud that efficiently handles large-scale heterogeneous datasets for real-time insights and predictive analytics in logistics.

Business Problem	Solution	Outcome
Cannot predict delivery delays	XGBoost delay prediction	AUC 73.1%, Recall 56.8%
No fleet health visibility	Tier-aware anomaly detection	32,763 anomalies, 94.5% healthy
No real-time data pipeline	End-to-end AWS pipeline	1.8M records, Airflow orchestrated
No monitoring or alerting	CloudWatch + SNS	5 alarms, live dashboard

3. System Architecture

The solution follows a 9-layer Lambda architecture pattern, separating batch and streaming concerns across three S3 data zones with Apache Airflow orchestration.

Layer	Component	Technology
1. Data Sources	IoT sensors, bookings, weather	Python simulators
2. Ingestion	5 simulators + manifest logger	boto3, AWS S3
3. Raw Storage	globetrack-raw-data-lake	AWS S3 JSON and CSV
4. ETL Processing	5 ETL pipelines	Pandas, PyArrow
5. Processed Storage	globetrack-processed-data	AWS S3 Parquet Snappy
6. ML and Analytics	3 ML models + anomaly detection	Scikit-learn, XGBoost, LightGBM,Random Forest
7. Analytics Zone	globetrack-analytics-zone	AWS S3 Parquet
8. Orchestration	14-task Airflow DAG	Apache Airflow 3.1.8
9. Monitoring	5 alarms + dashboard	AWS CloudWatch + SNS

4. Data Sources and Ingestion

Three datasets ,each of which is architected to scale to 50GB+ in production environments. The current 1.8M record synthetic dataset demonstrates the full pipeline capability.

Dataset	Type	Records	Key Features
Vehicle IoT Sensors	Telemetry	600,000	GPS, fuel, engine, speed, 3 vehicle tiers
Customer Bookings	Operational Logs	600,000	Orders, cities, cargo, 8 delay rules
Weather and Traffic	External Feeds	600,000	8 cities, climate patterns, congestion, currency

Vehicle Tier Design

Tier	Vehicles	Fuel Efficiency	Engine Temp	Fault Probability
OLD	GT-001 to GT-017	4 to 6.5 kmpl	95 to 108C	25%
MID	GT-018 to GT-035	7 to 10 kmpl	82 to 94C	10%
NEW	GT-036 to GT-050	11 to 15 kmpl	70 to 81C	3%

Delay Rules Injected

Rule	Condition	Delay Probability
1	Thunderstorm or Fog weather	90%
2	Heavy Rain	75%
3	Long haul over 500km + bad weather	85%
4	Express + very heavy cargo over 9000kg	80%
5	Perishables + long haul over 400km	70%
6	Short haul under 200km + clear weather	92% delivered
7	Pharmaceuticals priority	75% delivered
8	Base distribution	45% delivered, 25% transit

5. ETL Processing Pipeline

Pipeline	Input	Output	Key Features Engineered
etl_iot_sensor.py	50 JSON files	Parquet 28 cols	fuel_efficiency_kmpl, engine_health_score, alert_severity
etl_booking_logs.py	50 JSON files	Parquet 28 cols	is_delayed, revenue_per_km, weather_distance_risk
etl_weather.py	40 JSON files	Parquet 25 cols	weather_severity, combined_risk_score, travel_advisory
etl_enrichment.py	3 Parquet files	2 enriched Parquets	IoT + weather join, bookings + weather join
etl_large_scale.py	120 CSV chunks	12 Parquet chunks	1.18M records, 6 chunks per dataset

6. Machine Learning Models

6.1 Delivery Delay Prediction

Three algorithms trained on 580,107 booking records.

Model	Accuracy	AUC	Recall	Logistics Score
XGBoost	69.0%	73.1%	56.8%	Best (0.4xAUC + 0.6xRecall)
LightGBM	70.8%	72.9%	52.3%	Runner up
Random Forest	69.4%	72.8%	55.5%	Feature importance

Key ML Engineering Decisions

Decision	Implementation	Impact
Model selection metric	0.4 x AUC + 0.6 x Recall	Logistics-focused, catches more delays
No data leakage	Split before balance, no weather features	AUC 73%
Decision threshold	0.45 instead of default 0.5	Better recall
Class imbalance	scale_pos_weight, is_unbalance, balanced_subsample	Each model handles imbalance correctly
Early stopping	10 rounds on XGBoost and LightGBM	Prevents overfitting on 580K rows
Interaction features	cargo_stress_index, distance_cargo_risk, express_risk_score	Captures dependencies

6.2 IoT Anomaly Detection

Tier	Model	Contamination	Anomalies	Rate
OLD trucks	Isolation Forest + LOF	9%	18,376	9.0%
MID trucks	Isolation Forest + LOF	5%	10,783	5.0%
NEW trucks	Isolation Forest + LOF	2%	3,604	2.0%

Key finding: Anomalous vehicles average 82.76 km/h vs 72.58 km/h normal — overspeeding is the primary anomaly pattern. Tier-aware detection prevents NEW trucks from being misclassified due to their different performance profile.

7. Deployment and Orchestration

7.1 Apache Airflow DAG

14-task DAG scheduled daily at 6AM IST orchestrates the complete pipeline from ingestion through ML to alerting.

Stage	Tasks	Dependency
Stage 1 Ingestion	3 simulators parallel	After start
Stage 2 Manifest	1 manifest logger	After ingestion
Stage 3 Large Gen	1 large dataset generator	After manifest
Stage 4 ETL	3 ETL pipelines parallel	After large gen
Stage 5 Enrichment	1 enrichment pipeline	After ETL
Stage 6 Large ETL	1 large scale ETL	After enrichment
Stage 7 ML	2 ML models parallel	After large ETL
Stage 8 Analytics	1 analytics	After ML
Stage 9 Alert	1 success notification	After analytics

8. Testing and Quality Assurance

Test Suite	Cases	Result
Suite 1 S3 Infrastructure	6	6/6 PASSED
Suite 2 Data Ingestion	4	4/4 PASSED
Suite 3 ETL Processing	5	5/5 PASSED
Suite 4 ML Models	7	7/7 PASSED
Suite 5 Analytics and Reporting	5	5/5 PASSED
Total	27	27/27

9. Key Business Insights

Insight	Finding	Business Action
Fleet health	94.5% healthy, OLD trucks 9% anomaly rate	Schedule maintenance for GT-001 to GT-017
Delay drivers	Thunderstorm and Fog cause 90% delays	Avoid scheduling in severe weather
Express risk	EXPRESS orders 53.94% delayed	Review Express commitments
Revenue	Rs 6.43 Billion across 580,107 orders	Mumbai and Pune highest revenue cities
Overspeeding	Anomalous vehicles avg 82.76 km/h	Implement speed policy
Night shift	Night vehicles avg higher speed	Enhanced monitoring 8PM to 5AM

10. Conclusion

This project demonstrates a complete End-to-End Cloud Big Data Solution for Real-Time Logistics Intelligence. The system ingests 1.8 million records from three heterogeneous data sources, processes them through 5 ETL pipelines, trains 3 machine learning models and delivers actionable insights through a Power BI dashboard — all orchestrated via Apache Airflow and monitored through AWS CloudWatch.

11. Tech Stack

Category	Technology	Version
Cloud	AWS S3, Glue, Athena, CloudWatch, SNS	Latest
Language	Python	3.12.7
ETL	Pandas, PyArrow, Parquet Snappy	Latest
ML	Scikit-learn, XGBoost, LightGBM	Latest
Orchestration	Apache Airflow	3.1.8
Monitoring	AWS CloudWatch + SNS	Latest
Visualisation	Power BI Desktop	Latest
Version Control	Git + GitHub	2.53.0